



Bacterial Diarrhea — *E. coli*, *Shigella*, *Salmonella*, *Campylobacter*

Microbiology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

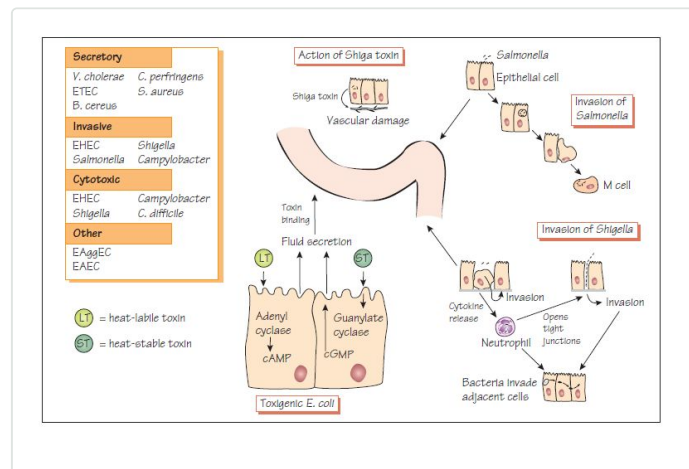
1. Which stool test finding tells you a diarrhea is invasive rather than toxin-mediated?
2. Which *E. coli* pathotype produces heat-stable / heat-labile toxins, and which causes haemolytic uraemic syndrome?
3. How do *Salmonella* and *Shigella* differ on TSI agar, H₂S and motility?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Big picture — how bacteria cause diarrhea

- WHO definition: **≥3 loose or liquid stools per day** (or more than normal for that person); leading cause of morbidity and death in **children under 5**
- Host defences: **gastric pH, IgA, normal gut flora, peristalsis**. Acid is reduced by milk, **H₂ antagonists and PPIs**
- Two routes: **infection** (live pathogen) or **intoxication** (pre-formed toxin in food; live organisms not required). Spread is **faecal-oral**

	Secretory (toxin-mediated)	Dysenteric / invasive
Mechanism	Toxin dysregulates intestinal cells → fluid secretion	Invasion and destruction of the intestinal wall; colonic inflammation
Stool	Infrequent, large-volume watery stools	Frequent, often blood-stained stools (e.g. <i>Shigella</i>)
Methylene-blue stool	No neutrophils: <i>V. cholerae</i> , <i>E. coli</i> , <i>C. perfringens</i>	Neutrophils : <i>Shigella</i> , <i>Salmonella</i> , <i>Campylobacter</i>



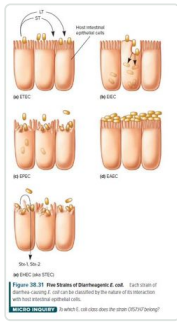
Secretory / invasive / cytotoxic pathogens; LT → cAMP, ST → cGMP; *Shigella* and *Salmonella* invasion.

2 *E. coli* — five diarrhoeagenic pathotypes

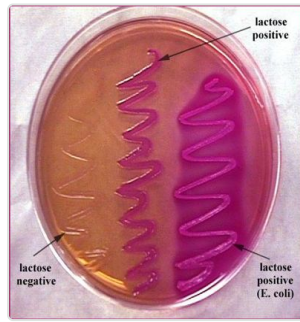
- Gram-negative bacillus, facultative anaerobe, **lactose fermenter** (pink on MacConkey; **green sheen on EMB**); part of normal colon flora; harmless strains become pathogenic by **horizontal gene transfer** (plasmids, prophages, pathogenicity islands)
- Serotyped by **O (LPS) and H (flagellar)** antigens; **pili** attach to jejunum / ileum; toxins are cell-specific — **colon cells lack the receptors**

Pathotype	Who / key disease	Mechanism / discriminator
EPEC	Infants in developing countries; severe watery diarrhea, vomiting, fever	Usually self-limited; can be prolonged or chronic
ETEC	Commonest cause of travellers' diarrhea ; children <5 in developing countries	ST → ↑cGMP; LT ≈ cholera toxin (↑cAMP) → hypersecretion, watery diarrhea
STEC (O157:H7)	Mild non-bloody diarrhea → haemorrhagic colitis → HUS (acute renal failure)	Shiga cytotoxin ; infectious dose <200 CFU ; cattle reservoir, undercooked beef; sorbitol non-fermenter

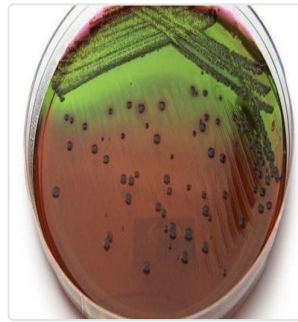
EIEC	Children in developing countries, travellers; shigellosis-like	Invades mucosal epithelium; non- / late-lactose fermenter, non-motile
EAEC	Acute and chronic (>14 days) diarrhea; travellers; persistent diarrhea in HIV	Characteristic aggregative adherence pattern



Five diarrhoeagenic *E. coli* strains.



Lac+ (pink) vs Lac- on MacConkey.



E. coli on EMB (green sheen).



O157:H7 does not ferment sorbitol.

Lab: Gram stain of stool is **not useful**; culture + immunoassay for Shiga toxin; process within 2 h or store at 4 °C. Treatment: rehydration; antibiotics usually not indicated.

3 Discriminator grid — *Shigella* vs *Salmonella* vs *Campylobacter*

	<i>Shigella</i>	<i>Salmonella</i> (non-typhoidal)	<i>Campylobacter jejuni</i>
Morphology / O₂	Gram-negative rods; facultative	Gram-negative bacilli; facultative	Curved / S-shaped; microaerophilic (5% O ₂); grows at 42 °C
Motile?	Non-motile	Motile	—
Lactose / H₂S / gas	Non-lactose fermenter (<i>S. sonnei</i> late); no H₂S, no gas	Non-lactose fermenter; H₂S +	Oxidase +; no growth at 25 °C
TSI agar	Alkaline slant / acid butt, no gas, no H₂S	Alkaline / acid, often gas + H₂S (black)	—
Infective dose	~10² (lowest) ; highly communicable	≥10⁵	As few as 500
Incubation	1–4 days	12–48 h	2–10 days
Picture	Fever, cramps → watery then bloody mucoid stool (dysentery); M-cell invasion, cell-to-cell spread, ulcers, "pseudomembrane"	Nausea, vomiting → pain → non-bloody diarrhea; neutrophils limit spread to gut + mesenteric nodes; bacteraemia infrequent	Watery, foul-smelling → bloody , fever, severe pain; recovery 5–8 days
Source	Faecal-oral; poor hygiene; bloodstream invasion rare	Contaminated food / water from animals (poultry, eggs); <i>S. typhimurium</i> commonest in US	Poultry, meat, unpasteurised milk , puppies; in the US enteritis exceeds <i>Salmonella</i> + <i>Shigella</i>
Special point	Species: <i>sonnei</i> , <i>flexneri</i> , <i>dysenteriae</i> , <i>boydii</i>	Typhoidal (enteric fever) vs non-typhoidal (enterocolitis)	Guillain-Barré by molecular mimicry
Treatment	Self-limited (5–7 d); oral fluids; avoid antidiarrhoeals ; antibiotics for severe cases	Usually self-limited; antibiotics do not shorten illness, prolong excretion , raise carrier state, select resistance; only for neonates / chronic disease at risk of septicemia	Fluids; erythromycin, fluoroquinolones ; penicillin-resistant

4 Lab media — quick identification

Organism	Medium result
<i>E. coli</i>	Pink colonies on MacConkey; green sheen on EMB ; O157 = colourless on sorbitol MacConkey
<i>Shigella</i>	Colourless (non-lactose fermenting) on MacConkey / EMB; Hektoen Enteric; confirm by slide agglutination
<i>Salmonella</i>	Colourless on MacConkey / EMB; black colonies on Hektoen (H ₂ S + ferric ammonium citrate); blood culture for enteric fever (first 2 weeks) — stool for enterocolitis

Campylobacter

Skirrow agar, 42 °C, 5% O₂ + 10% CO₂; oxidase +, nalidixic-acid sensitive; poorly stained by safranin (use carbol fuchsin)



Shigella on Hektoen.



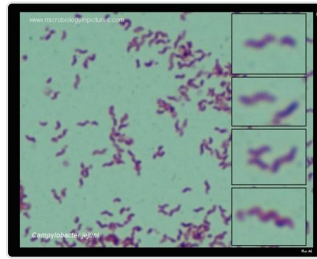
Shigella on TSI.



Salmonella on Hektoen.



Salmonella on TSI.



Campylobacter Gram stain.



Skirrow agar.

Prevention: proper food preparation (especially poultry), avoid unpasteurised dairy, protect water supplies.

P Must-remember pearls

- **Neutrophils in stool** = invasive (Shigella, Salmonella, Campylobacter); toxin producers (ETEC, cholera, C. perfringens) give watery stool without them.
- **ETEC** = travellers' diarrhea (LT/ST); **STEC O157:H7** = bloody diarrhea + **HUS**, **sorbitol non-fermenter**.
- **Shigella**: non-motile, no H₂S, lowest dose (~10²), bloody dysentery — **avoid antidiarrhoeals**. **Salmonella**: motile, H₂S +, dose ≥10⁵ — **no antibiotics** in usual enterocolitis.
- **Campylobacter**: curved, microaerophilic, **42 °C**, poultry; complication **Guillain-Barré**.
- Gastric acid is the key defence against Salmonella; PPIs / H₂ blockers weaken it.

Q Test yourself

1. Teenagers develop bloody diarrhea after undercooked hamburgers and two develop HUS. *E. coli* O157:H7 was isolated. The "O157" refers to which structure?

- A. Lipopolysaccharide
- B. Capsule
- C. Flagella
- D. Fimbriae

2. A hospitalised woman develops diarrhea five days into ceftriaxone; colonoscopy shows fibrin-and-WBC plaques; Salmonella, Shigella, Campylobacter and amoebae are negative. Most likely cause?

- A. Endotoxin from clostridial spores
- B. Exotoxin from clostridial cells
- C. Exotoxin from *E. coli* cells
- D. Endotoxin from *E. coli* cells

3. Which host defence is MOST important in preventing Salmonella infection?

- A. Gastric acid
- B. Salivary enzymes
- C. Oral flora
- D. Alpha interferon

4. Which organism most likely invades the intestinal mucosa?

- A. *Vibrio cholerae*
- B. *Shigella sonnei*
- C. Enterotoxigenic *E. coli*
- D. *Clostridium botulinum*

5. Regarding Salmonella food poisoning, which statement is true?

- A. Symptoms start within 3 hours
- B. Antibiotics are not indicated in most cases
- C. Food handlers are the main source
- D. Blood cultures are frequently positive

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — A** — O antigens are the polysaccharide part of LPS; H antigens are flagellar.
- 2 — B** — C. difficile toxins after antibiotics (this case is on the SGL 4 slides but the organism is taught in SGL 1).
- 3 — A** — The Salmonella dose is high (≥10⁵), so gastric acid is the main barrier.
- 4 — B** — Shigella invades epithelial cells; V. cholerae and ETEC are secretory toxin producers.
- 5 — B** — Usually self-limited; incubation is 12–48 h; diagnosis is usually by stool culture.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam